

Android Static Analysis Capability Catalog

Security & Privacy Signal Detection Reference

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Categories

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Detection Signals

01 Network Security & Transport Layer

SSL/TLS connection and cleartext HTTP usage detection	WebView SSL client configuration inspection
Custom SSL implementation inspection	Certificate pinning absence detection
Hostname verification in SSL configuration	SSL certificate validation correctness check
Unsafe SSL components usage detection	General network access behavior inspection
HttpHost insecure transport usage	

02 Sensitive Data Access & Device Information

Contact list access detection	Device identifiers (IMEI / Device ID) access detection
Call log access detection	Email address exposure detection
SMS message access detection	Media URI / file URI usage detection
Microphone recording access detection	URL / IP address string presence detection
Precise GPS location access detection	Sensitive data encoded in Base64 detection

03 Application Permissions & Manifest Configuration

AndroidManifest declared permissions inspection	ContentProvider exported exposure detection
Dangerous permission usage detection	Implicit service exposure detection
Normal permission usage inspection	Intent filter misconfiguration analysis
Permission group configuration analysis	SharedUserId usage detection
Exported component misconfiguration detection	ADB backup enabled configuration detection

04 WebView & Browser-Based Security

WebView insecure configuration detection	Deep link scheme / host validation analysis
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JavaScript interface injection (addJavascriptInterface) risk detection	Deep link injection vulnerability detection
WebView advanced security configuration inspection	

05 Cryptography & Key Management

Weak cryptographic algorithm usage detection	Cryptographic framework usage inspection (Bangle / Ijiami / MonoDroid)
Secure key storage (AndroidKeyStore) misuse detection	Key usage for database encryption detection
Hardcoded API keys or secrets detection	Weak hashing algorithm (MessageDigest) usage detection

06 Storage & File System Security

External storage usage detection	SQLite encryption implementation validation (SQLCipher / SEE)
Insecure file deletion handling detection	ZipSlip (path traversal in ZIP extraction) detection
SQLite database vulnerability detection	

07 Code Execution & Runtime Behavior

Runtime command execution detection	Unsafe class or method usage detection
Native library loading inspection	Security-related API usage inspection
Dynamic code loading detection	

08 Debugging, Tracking & Logging

Debug mode enabled detection	User tracking or identifier collection detection
Logging of sensitive data (Log.d / Log.v) detection	Installation source verification detection

09 System Security & Device Protection

Root detection implementation inspection	Biometric authentication implementation correctness
Screenshot prevention configuration detection	Tapjacking / overlay attack protection detection
App sandbox integrity validation	

10 Payment, Billing & External Services

Google Play Billing API usage detection	
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11 Frameworks & Third-Party Security Layers

Third-party encryption framework detection (Bangle / Ijiami)	MonoDroid framework usage detection
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